Top-Degree Rank $\dim_{\mathbb{Q}} R_{12}(F) = 1751$ for the Three-Chain Sextic Jacobian Algebra and Independence from Contained Planes

Benjamin Stanley Frohman
Independent Researcher | ORCID: 0009-0006-7068-3718
frohmanbenjamin@gmail.com

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Abstract

Let $W(u,v) = u^5v+v^6$ and let $F = W(x_0,x_3)+W(x_1,x_4)+W(x_2,x_5) \in \mathbb{C}[x_0,\dots,x_5]$ define the smooth three-chain sextic fourfold $X = V(F) \subset \mathbb{P}^5$ containing the plane $\Pi = V(x_3,x_4,x_5)$. We prove that containment of the plane $\Pi \subset X$ does not alter any graded dimension of the Jacobian ring $J(F) = \mathbb{C}[x_0,\dots,x_5]/(\partial F)$ relative to a very general smooth sextic fourfold: the partials form a regular sequence of six degree-5 forms, $\dim_{\mathbb{C}} J(F) = 5^6 = 25^3 = 15625$, the Hilbert series is $(1+t+t^2+t^3+t^4)^6$, and the degree-12 Griffiths residue piece has $\dim R_{12}(F) = 1751$ (so $h^{2,2}(X) = 1752$ and $b_4(X) = 2606$). We explicitly separate this graded Jacobian dimension from the rational Hodge rank $2 \le \rho_{\mathrm{Hdg}}(X) \le 1752$.

1 Main Theorem

Theorem 1 (Three-Chain Jacobian Graded Dimension)**.** *For $F = \sum_{i=0}^{2}(x_i^5x_{i+3} + x_{i+3}^6)$ on $\mathbb{P}^5$:*

1. *$V(\partial F) = \{0\} \subset \mathbb{C}^6$, so $X = V(F) \subset \mathbb{P}^5$ is smooth and $(\partial_0 F,\dots,\partial_5 F)$ is a regular sequence of homogeneous degree-5 forms.*
2. *$J(F) \cong J(W)^{\otimes 3}$ has total dimension $\dim_{\mathbb{C}} J(F) = 25^3 = 5^6 = 15625$ and Hilbert series $\sum_{k=0}^{24}(\dim R_k)t^k = (1+t+t^2+t^3+t^4)^6$.*
3. *By Griffiths residue isomorphism, $h^{4,0} = 1$, $h^{3,1} = \dim R_6 = 426$, $h^{2,2}_{\mathrm{prim}} = \dim R_{12} = 1751$, and $h^{2,2}(X) = 1752$, independent of the contained plane $\Pi = V(x_3,x_4,x_5) \subset X$.*

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